Data Example: Phosphorous Levels

Phosphorous Example A – Standard Linear Regression

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1 Phosphorous Example A: Standard Linear Regression

In this implementation, we will be using phosphorous example A from lecture to explore standard linear regression.

1.1 Phosphorous A

Inorganic P (ppm) *(explanatory variable)*			
0.7	1.2	3.6	19.0
1.0	0.8	18.5	27.8
1.4	0.9	21.9	20.9

Organic P (ppm) *(response variable)*			
0.1	1.9	5.1	26.5
1.8	2.0	26.3	38.6
2.3	1.3	30.1	28.4

2 Ordinary Least Squares

We have seen evidence of spatial correlation in the inorganic and organic phosphorous levels. What about the relationship between phosphorous levels in the inorganic and in the organic? We start with the standard linear regression without any spatial consideration.

For this, the response variable is organic phosphorous levels in the corn and the explanatory variable is inorganic phosphorous levels in the soil. The data are stored in the data frame, `phosA`. We use the `lm()` function to fit a simple linear regression and save it as an `lm` object called `phos.lm`.

```
library(spdep)
library(spatialreg)
library(ggplot2)
library(MASS)

phos.inorganic <- c(0.7, 1.2, 3.6, 19.0,
                    1.0, 0.8, 18.5, 27.8,
                    1.4, 0.9, 21.9, 20.9)

phos.organic <- c(0.1, 1.9, 5.1, 26.5,
                  1.8, 2.0, 26.3, 38.6,
                  2.3, 1.3, 30.1, 28.4)

plot <- as.factor(seq(1, 12))
```

```r
phosA <- cbind.data.frame(phos.inorganic, phos.organic, plot)

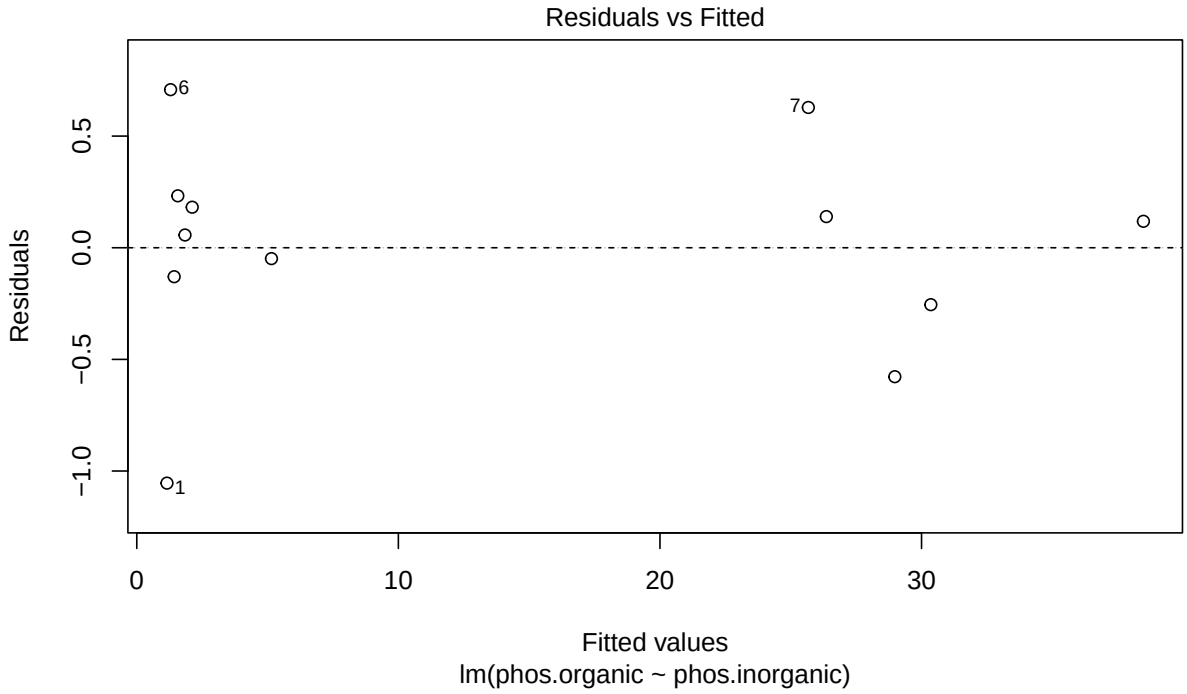
phos.lm <- lm(phos.organic ~ phos.inorganic, data = phosA)
summary(phos.lm)
```

```
##
## Call:
## lm(formula = phos.organic ~ phos.inorganic, data = phosA)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.05426 -0.16104  0.08774  0.19431  0.70800
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     0.19008    0.20185   0.942    0.369
## phos.inorganic  1.37739    0.01424  96.742 3.41e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5049 on 10 degrees of freedom
## Multiple R-squared:  0.9989, Adjusted R-squared:  0.9988
## F-statistic:  9359 on 1 and 10 DF,  p-value: 3.411e-16
```

From the model summary output above, we find that the fitted regression coefficient on inorganic phosphorous is `1.37739`, indicating that an increase in 1 ppm of inorganic phosphorous is expected to be associated with an increase of `1.37739` ppm in organic phosphorous. The standard error is `0.01424`, the t-value is `96.742`, and the p-value is `3.41 \times 10^{-16}`. There is very strong evidence of a relationship between inorganic phosphorous and the organic phosphorous.

Next, we perform diagnostics of this model by residual plots. We apply the `plot()` function to `phos.lm`. We specify `which=1` and `which=2` in order to only plot one plot at a time (the first two default plots). `plot()` with `which=1` gives us the usual residual plot of residuals versus the fitted values of the organic phosphorous. We also specify the argument `add.smooth=FALSE` in order to omit the default localized smoothing line. Instead, we superimpose a zero horizontal line to the residual plot by the `abline()` function.

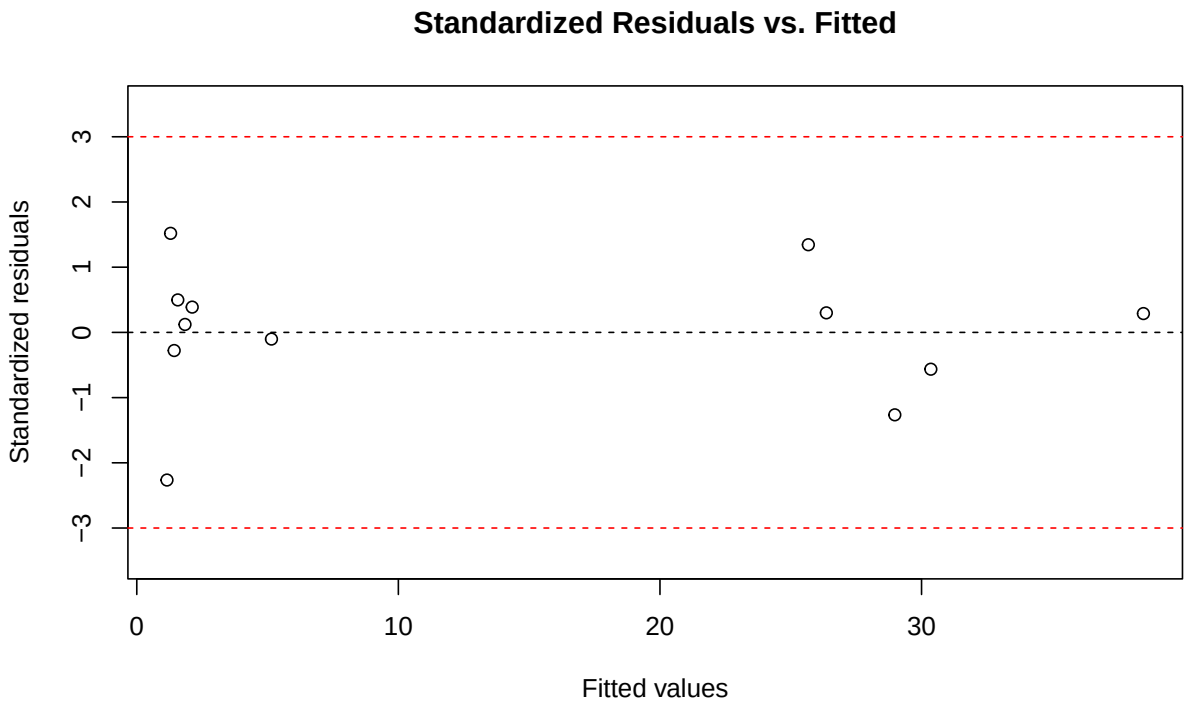
```r
# plot residuals vs. fitted
plot(phos.lm, which = 1, add.smooth = FALSE)
abline(h = 0, lty = 2)
```



What do we see in the residual vs. fitted plot? First, we see the residuals are somewhat scattered around the zero line. However, the pattern is somewhat obscured by an unusually large negative residual (observation 1, flagged automatically by R, in addition to observations 6 and 7). Thus, we will plot the standardized residuals next. The standard deviation for residuals in a linear model can vary a lot across observations. Therefore, by standardizing the residuals, we can make comparisons that are more useful.

In order to plot the standardized residuals vs. fitted values, we can use the `stdres` function from the `MASS` package. Here, we also plot the horizontal lines indicating at negative 3 and positive 3. A rule-of-thumb is to declare an outlier if the standardized residual is greater than 3 or less than negative 3.

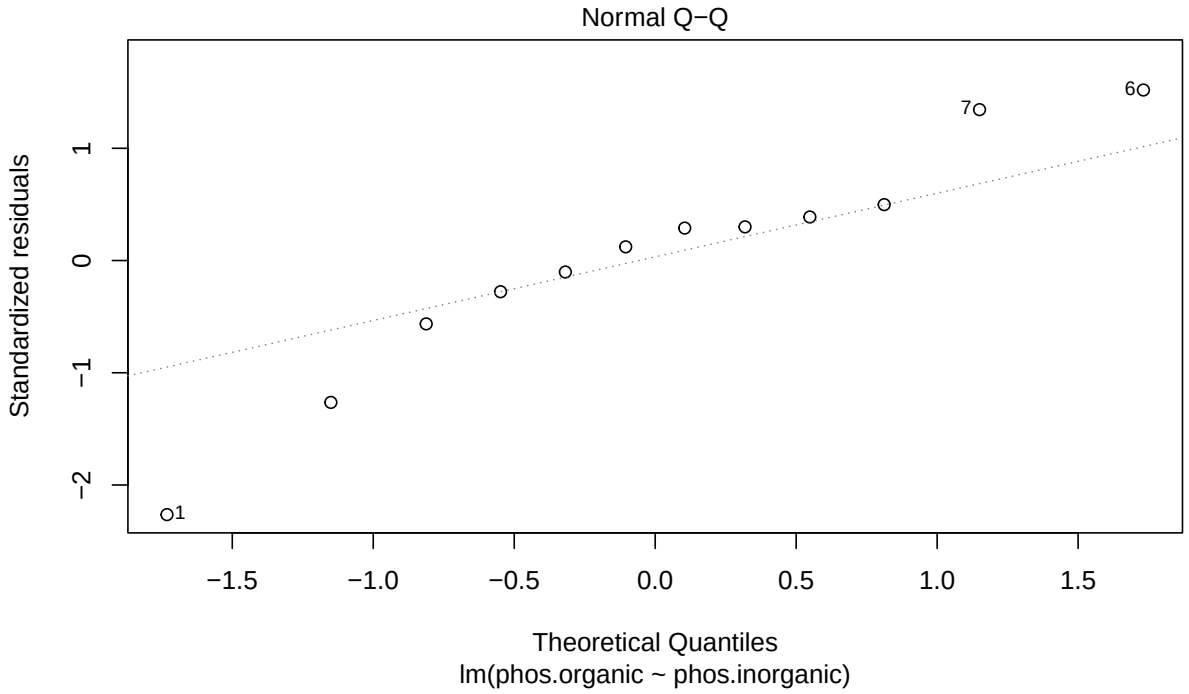
```
# plot 3s standardized residuals to explore potential outliers
stdresids <- stdres(phos.lm)
plot(x = fitted(phos.lm), y = stdresids,
     main = "Standardized Residuals vs. Fitted",
     ylim = c(-3.5, 3.5),
     xlab = "Fitted values",
     ylab = "Standardized residuals")
abline(h = c(-3, 0, 3), col = c("red", "black", "red"), lty = 2)
```



From this plot, we see a random scatter of the standardized residuals around zero indicating no obvious departure from the model assumptions. There is also no indication of any outlier.

`plot()` with `which=2` gives us the normal quantile-quantile (or Q-Q) plot of the standardized residuals.

```
# plot QQ plot
plot(phos.lm, which = 2, add.smooth = FALSE)
```



From the Q-Q plot above, we observe departure from the 45 degree line at both ends, indicating somewhat heavy tails on either end of the distribution. However, it is difficult to judge normality with only 12 observations.

3 Testing Residuals Using Moran's I

To evaluate the independence assumption, we test for spatial patterns in the residuals using the function `moran.test()`.

First, we recreate the rook neighborhood structure and the binary spatial weights.

```
phos.nb <- cell2nb(nrow = 3, ncol = 4, type = "rook")
phos.lw <- nb2listw(phos.nb, style = "B")
```

Now test spatial correlation in the residuals.

```
# Test spatial correlation in residuals
set.seed(3)
moran.test(resid(phos.lm), phos.lw, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(phos.lm)
## weights: phos.lw
##
## Moran I statistic standard deviate = 0.84983, p-value = 0.3954
```

```
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.07583167       -0.09090909        0.03849589
```

From the output above, the observed Moran's I statistic is `0.07583167` and the p-value is `0.3954`. We do not reject the null of CSR at the 10% level. There is no evidence against the null hypothesis of CSR.